

Human Resource Management (HRM) System

Course Code: Practicum CSC 490

Prepared for:

Practicum Defense Committee



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Introduction

- **Problem:** small and medium organizations still track attendance on paper or spreadsheets, handle leave by e-mail, and calculate salaries by hand – slow, error-prone and opaque to employees.
- **Solution:** a web-based, **multi-company HRM System** that automates attendance, leave and monthly payroll with role-based access.
- **Roles:** Super Admin (onboards companies), Company Admin (manages own company), Common employee (checks in/out, applies for leave, views own records).
- **Automation:** a daily job marks absent employees; a monthly job generates the previous month's payroll.
- **AI HR Assistant:** answers HR questions in natural language from the company's own data.

Objectives

- **Broad objective**
 - Design and develop a secure, web-based, multi-company HRM system that automates attendance, leave and payroll administration.
- **Specific objectives**
 - Role-based access control scoped to the user's own company
 - Check-in/check-out with automatic late and early-leave detection; scheduled absent marking
 - Leave workflow: request → approve/reject → reflected in attendance
 - Automatic monthly payroll with deduction rules; live payroll projection
 - Daily, monthly and per-employee attendance analytics
 - AI assistant using read-only, company-scoped data tools
 - JWT authentication, refresh-token rotation, BCrypt password hashing



Organization Overview

- **Organization:** AKIJ iBOS Ltd., Dhaka, Bangladesh
- **Services:** Provides AI-driven software, enterprise solutions, and full-stack digital services designed for modern businesses
- **Vision / Mission:** Vision is to build smart systems
- **My position:** Backend Intern – analysis, design and full-stack development of the HRM System
- **Organizational supervisor:** Saidur Rahman Akash, Software Engineer L-2
- **Duration:** 12 weeks (1st June, 2026 – 31st August, 2026)



Methodology and Process Model

- **Process model:** Agile – iterative and incremental with feedback at the end of every iteration.
- **Iterations (12 weeks)**
 - Sprint 0 – requirements, Clean Architecture skeleton, database schema
 - Sprint 1 – JWT authentication, company / role / department management
 - Sprint 2 – employee management, pagination, Swagger security
 - Sprint 3 – attendance, automatic absent marking, leave workflow
 - Sprint 4 – payroll job, live payroll status, AI assistant
 - Final – Angular integration, testing, documentation
- **Why Agile:** requirements were refined from working software (e.g. the check-in rule after auto-absent marking), modules could be demonstrated early, defects found early.

System Requirements

Role	Key functional requirements
Super Admin	Create, update, activate/deactivate and list companies (paginated)
Company Admin	Manage departments (with employee counts), roles and employees; view attendance by date/employee and daily/monthly summaries; approve, reject or cancel leave; view payroll and live projection; ask the AI assistant
Common (Employee)	Log in; check in / check out once per day; submit leave requests; view own attendance, statistics and payroll
System (Scheduler)	Mark absent employees every working day at 09:30 (skip holidays and approved leave, catch up at startup); generate previous month's payroll on the 1st, exactly once

- **Non-functional:** JWT security and company-level data isolation; paginated lists (default 50, max 100); idempotent jobs; responsive Angular UI; Clean Architecture + CQRS for maintainability.

Use Case Diagram

Human Resource Management System with AI Bot

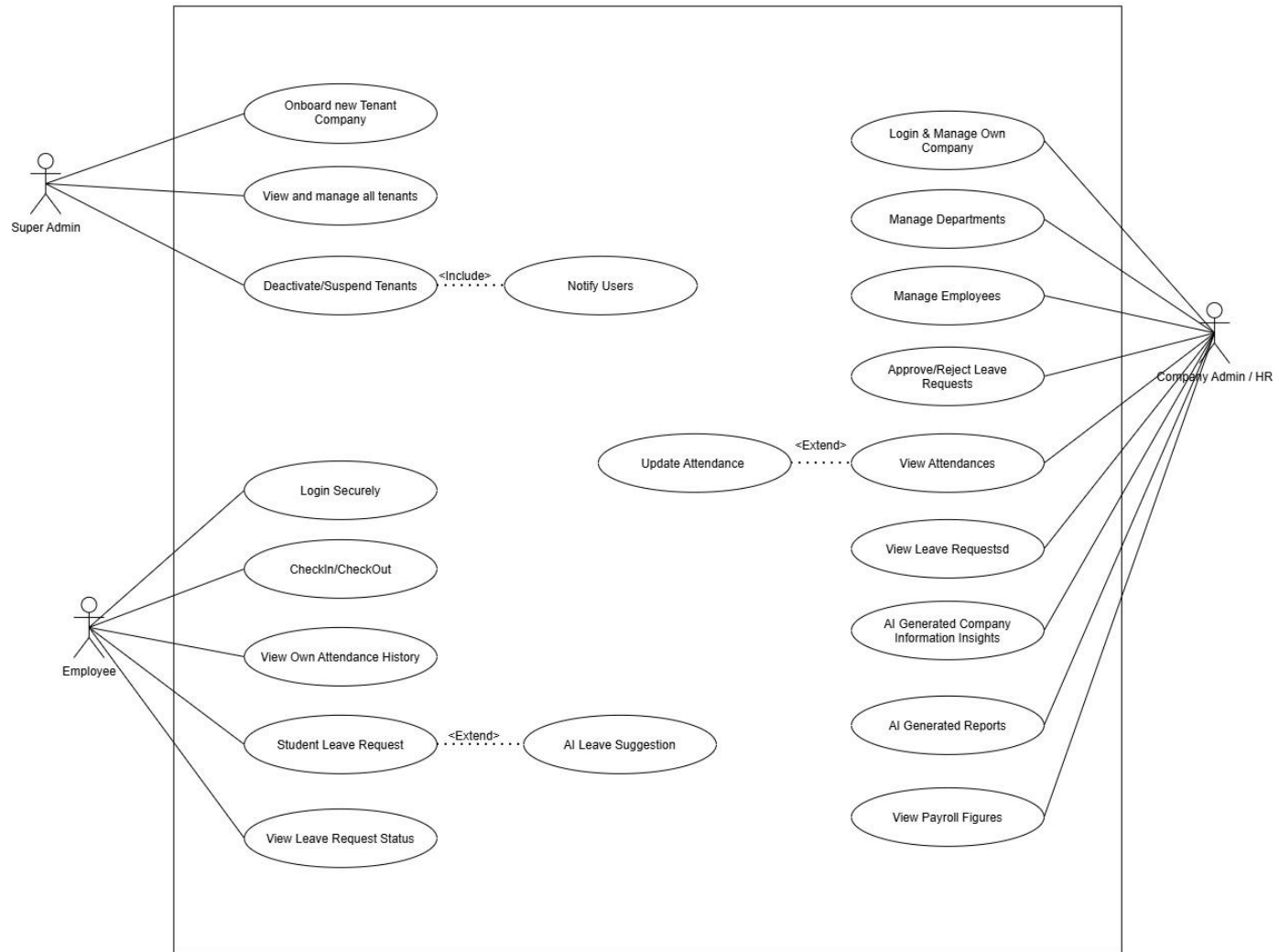
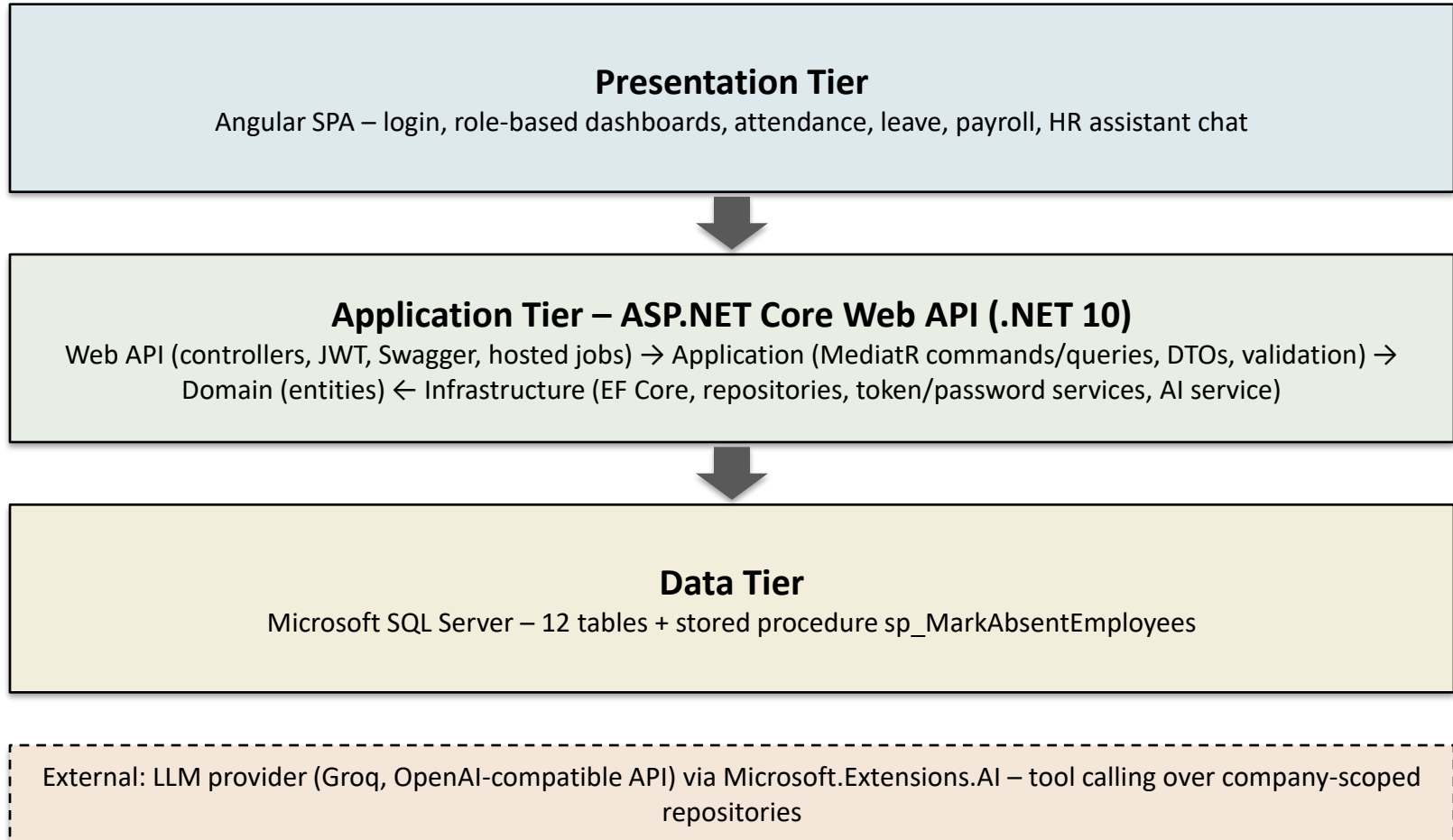


Fig : Use Case Diagram

Figure: Use case diagram of the HRM System (three primary actors, Scheduler as secondary actor; «include» Mark On-Leave Days, «extend» Convert Absent to Late).

System Architecture



- **Pattern:** Three-tier client–server with **Clean Architecture** and **CQRS (MediatR)** – every use case is an isolated command/query handler.
- **Security:** HS256 JWT (15 min) + HttpOnly refresh cookie (7 days); company id read only from the token.

Database Design

ERD – HR Management System with AI Bot

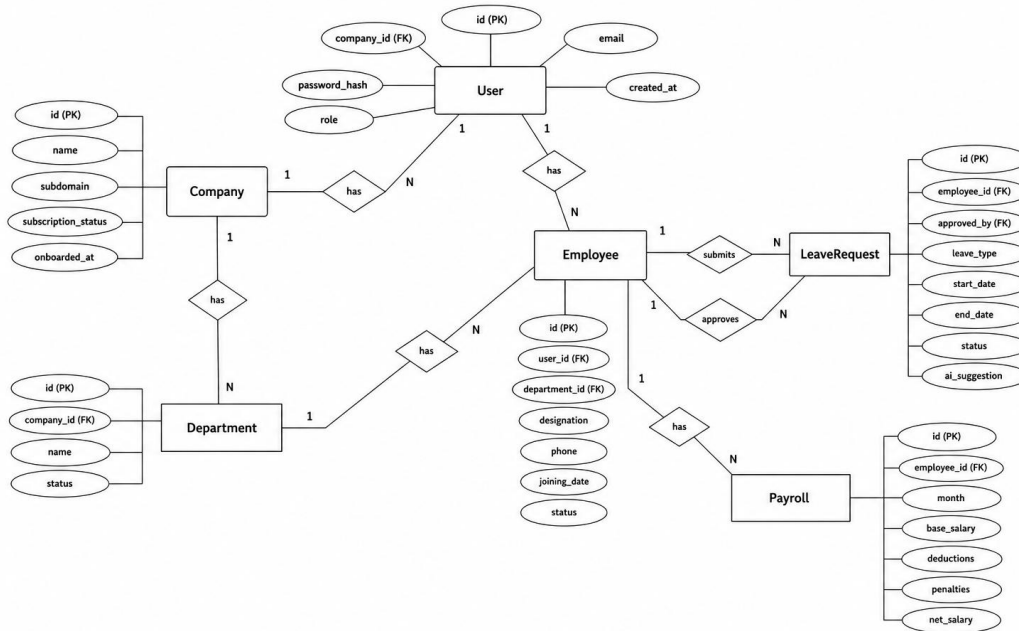


Table	Purpose
Company	Tenant organizations
Department, Role	Company structure, authorization roles
Employee	Users: profile, credentials, salary
Attendance	One row per employee per day
LeaveType, LeaveRequest	Leave workflow
Payroll	Monthly salary with deductions
RefreshToken	Session renewal / revocation
AiConversation, AiAuditLog, AiReport	AI usage and audit

- 12 normalized tables; every business table carries **CompanyId** for tenant isolation.
- Attendance status: Present / Late / Absent / On Leave; Payroll: Basic – AbsentDeduction – LateDeduction = Net.



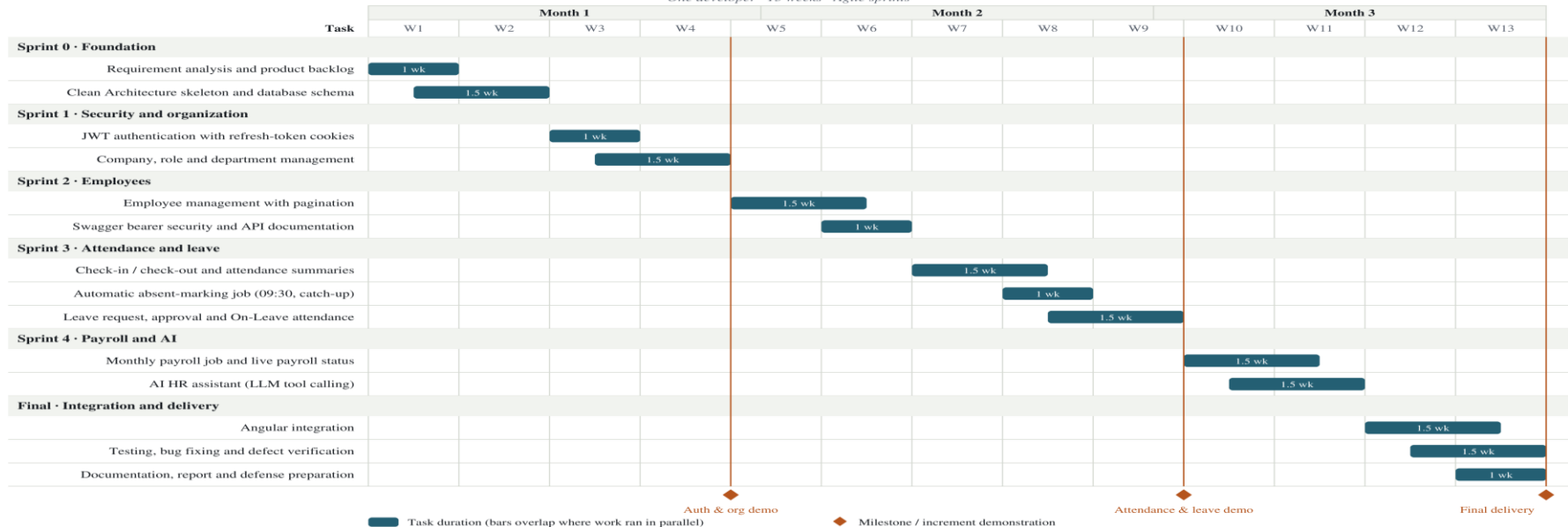
Project Estimation and Schedule

Metric	Value
Transactional UFP (14 EI + 7 EO + 11 EQ)	$59 + 36 + 41 = 136$
Data UFP (12 ILF $\times$ 7)	84
Total UFP	220
$VAF = 0.65 + 0.01 \times TDI$ (41)	1.06
Adjusted Function Points	233.2 FP
Actual effort (1 dev $\times$ 3 months $\times$ 22 days $\times$ 8 h)	528 person-hours $\approx$ 2.26 h/FP

Cost item	BDT
Personnel (528 h $\times$ 227.27)	120,000.00
Hardware (depreciation)	3,083.33
Other (internet, power, backup)	12,450.00
Software (all open source)	0.00
Total	135,533.33

HRM System – Project Schedule (Gantt Chart)

One developer · 13 weeks · Agile sprints



Salary and hardware figures are illustrative assumptions (BDT 40,000/month, laptop BDT 85,000).

Risk Management (RMMM)

Risk	P / I	Mitigation	Contingency
Unauthorized or cross-company access	Mod / Cat	JWT + roles, company id from token, BCrypt, refresh rotation	Revoke tokens, rotate key, fix
Incorrect attendance from auto-absent job	High / Ser	Idempotent SP, holiday calendar, startup catch-up, Absent→Late conversion	Admin corrects; re-run job
Duplicate / wrong payroll	Mod / Cat	Existence check; single formula for generation and projection	Regenerate after fixing data
LLM provider failure or wrong answer	High / Tol	Configurable model; read-only scoped tools; toolsUsed returned	Switch model / disable
Database or server failure	Mod / Ser	Backups, transactions, monitoring	Restore backup
User operation error	Mod / Tol	Validation, confirmations, training	Assisted correction

P = probability, I = impact (Tolerable / Serious / Catastrophic). Risks 2 and 4 were actually encountered during development and resolved.



Implementation – Screenshots

H

HRM

Email

Password

Log in

[Log In Page](#)

Search

Sahadat · Company Admin

Employees

+ New Employee

Department

All departments

Name	Email	Role	Status	
Mash Tarafder	mashrafeiubat@gmail.com	Super Admin	Active	Edit
Sahadat Sanbid	sahadat@s.s	Company Admin	Active	Edit
Din Islam	r@r.i	Common	Active	Edit
Mash T	mash@m.m	Common	Active	Edit
rakin a	r@r.i	Common	Active	Edit

Search

Sahadat · Company Admin

Dashboard

My Company

View and edit your company's details.

Go to My Company

Departments

Manage departments per company.

Go to Departments

Employees

Browse and manage employee records.

Go to Employees

Leave Approvals

Review pending leave requests.

Go to Approvals

Attendance

View attendance for a company and date.

Go to Attendance

Payroll

View payroll across a company.

Go to Payroll

Search

Sahadat - Company Admin

Attendance

Today

Check-in: —

Check-out: —

Check In

Check Out

This Month

2

Present

2

Late

0

Leave

20%

Ratio

Date	Check In	Check Out	Status
2026-07-18	18:23:14.7323355	—	Late
2026-07-17	18:27:28.3173534	—	Late
2026-07-16	00:55:54.2961033	10:34:01.2765384	Present
2026-07-20	—	—	Absent
2026-09-04	22:40:06.9885406	22:40:20.4430826	Late

Angular frontend consuming the Web API; role decides the dashboard after login.



Implementation – Screenshots (Cont.)

Search

Sahadat - Company Admin

My Leave Requests

+ New Request

From	To	Days	Reason	Status
2026-09-15T00:00:00	2026-09-20T00:00:00	6	123456	Approved

Search

Sahadat - Company Admin

Attendance — Monthly Summary

Day Month

Department: All departments Month: September Year: 2026

9% Avg. Attendance

2 Late Arrivals

0 Perfect Attendance

HR Most Punctual Dept.

Highest absentee this month: Din Islam, Mash T, Mash Tarafder, rakin a - Late rate: 9.09%

Employee	Department	Total Absent
Din Islam	HR	3
Mash T	HR	3
Mash Tarafder	HR	3
rakin a	HR	3
Sahadat Sanbid	HR	2

Search

Sahadat - Company Admin

Payroll — Company

Month: 9 Year: 2026

Employee	Employee Id	Basic Salary	Absent Deduction	Late Deduction	Net Salary
No payroll records for this period.					

Live Estimate (Current Month)

Computed on demand from this month's attendance so far — not a finalized/generated payroll record.

Employee	Basic Salary	Absent Deduction	Late Deduction	Net Salary
Mash Tarafder	7000	300	0	6700
Sahadat Sanbid				
Din Islam				

HR Assistant

Ask about attendance and leave in plain language. Each question is answered on its own.

You: Who was absent today?

Assistant: **Absent today (2026-09-14):**

- Mash Tarafder
- Sahadat Sanbid
- Din Islam

Ask a question about attendance or leave...

0 / 1000

Send

Payroll rule: 100 BDT per absence; every 3 late arrivals = 1 absence. AI answers list the data tools used.

Testing

- **Approach:** black-box (Swagger + Angular UI), white-box (handlers, branches), grey-box (DB verification); unit → integration → system → acceptance.

Scenario	Expected	Result
Login and token refresh	Token with Role & CompanyId; refresh only with valid cookie	Pass
Role-based, company-scoped access	401 without token, 403 wrong role, no cross-company data	Pass
Check-in after auto-absent marking	Absent row converted to Late, no duplicate	Pass (after fix)
Payroll idempotency	Exactly one row per employee per month	Pass
AI question “Who was late the most?”	Correct name + count; toolsUsed listed	Pass

- **Non-functional:** 15-min token expiry, cookie flags, pagination limits, AI response time (2–10 s), Windows/Linux portability.

Limitations and Future Plan

- **Limitations**

- Shift times, holidays and penalty are global constants, not per-company settings
- Payroll covers basic salary and deductions only; generated globally
- AI assistant: attendance and leave only, no payroll, no conversation memory
- No notifications; no dedicated mobile app

- **Future plan**

- Per-company configuration of shifts, holidays, leave balances and deductions
- Allowances, tax, bonus and payslip export (PDF/Excel)
- AI: payroll tools, memory, proactive monthly reports
- E-mail/push notifications; mobile app with location/biometric check-in
- Automated test suites with CI/CD

Conclusion

- Delivered a working, secure, multi-company HRM System: attendance, leave, payroll, analytics and an AI assistant.
- Automation removes manual registers and hand-calculated salaries; company-level isolation is enforced by design.
- Applied the full lifecycle – requirements, Clean Architecture/CQRS design, FP estimation, RMMM, testing.
- Practicum value: hands-on experience with Angular, .NET 10, EF Core, JWT security, background services and LLM tool-calling, plus documentation and communication skills.
- The system is a solid base for the per-company configuration and payroll extensions planned next.

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NOTE: consult your supervisor before and after preparing the final defense slides.

Any Questions ?

THANK YOU